



Tech Interview Preparation Checklist

1) DSA Round

- **What:** Core problem-solving, data structures, algorithms.
 - **Format:** 1–2 problems, 45–60 mins, whiteboard or online IDE.
 - **How to Answer:**
 - Start with brute force → optimize.
 - State complexity before/after coding.
 - Dry-run + edge cases.
 - **Red Flags:** Jumping to code, no explanation, no complexity analysis.
 - **Examples:** Google, Microsoft, Amazon, Meta, Uber, Bloomberg, Apple, Adobe.
 - **Action:**  1 timed mock daily + post-mortem.
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2) Quant / Puzzle Round

- **What:** Math reasoning, probability, estimation, mental agility.
 - **Format:** Rapid-fire questions, pen & paper or oral.
 - **How to Answer:**
 - State assumptions clearly.
 - Structure > speed.
 - Sanity-check extreme cases.
 - **Red Flags:** Formula dumping, no assumptions.
 - **Examples:** Jane Street, HRT, IMC, Optiver, Akuna, Citadel.
 - **Action:**  Daily puzzle/EV problem.
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3) Project Teardown

- **What:** Ownership, depth, design decisions, trade-offs.
- **Format:** 30–45 mins, project/internship discussion.
- **How to Answer (S.O.D.A.):**
 - Scope, Options, Decisions, Aftermath.
 - Bring diagrams, metrics, demos.

- **Red Flags:** Saying “team project” with no personal input.
 - **Examples:** Stripe, Uber, Atlassian, Microsoft, Google.
 - **Action:** ^{July 4}₁₇ Prepare 2 projects with S.O.D.A. notes.
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4) Systems / Machine Coding

- **What:** Engineering craft—build structured mini systems.
 - **Format:** 60–120 mins (parking lot, rate limiter, newsfeed).
 - **How to Answer:**
 - Clarify requirements first.
 - Design data model & APIs.
 - Implement core → edges → tests.
 - **Red Flags:** Spaghetti code, no tests, no README.
 - **Examples:** Stripe, Uber, Flipkart, Swiggy, Walmart.
 - **Action:** ^{July 4}₁₇ Weekly build practice (1 system end-to-end).
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5) LLD (Low-Level Design)

- **What:** OO design, extensibility, modeling.
 - **Format:** Class design (ride matching, notifications).
 - **How to Answer (R.U.M.S.):**
 - Requirements, Use-cases, Modeling, Scenarios.
 - **Red Flags:** God classes, no interfaces, tight coupling.
 - **Examples:** Amazon, Uber, Flipkart, Atlassian, Adobe.
 - **Action:** ^{July 4}₁₇ 2 LLD mocks per week.
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6) HLD (High-Level Design)

- **What:** Architecture, scalability, trade-offs.
- **Format:** System design (URL shortener, chat, feed).
- **How to Answer:**
 - Clarify scale (QPS, latency, data size).
 - Core components (APIs, DB, cache, queues).
 - Add observability + resilience.
- **Red Flags:** Buzzwords, no numbers, ignoring bottlenecks.

- **Examples:** Google, Meta, LinkedIn, Stripe, Airbnb.
 - **Action:**  1 HLD case study per week.
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7) Hiring Manager Round

- **What:** Team fit, ownership, prioritization, leadership.
 - **Format:** Conversational. STAR + Impact stories.
 - **How to Answer:**
 - Situation, Task, Action, Result (+ Impact).
 - Ask team culture/north star questions.
 - **Red Flags:** Blame game, vague outcomes, no curiosity.
 - **Examples:** Amazon, Microsoft, Google, Meta, Netflix.
 - **Action:**  Write 5 STAR stories (impact quantified).
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8) HR Round

- **What:** Culture, motivation, logistics, compensation.
 - **Format:** Values, conflict, career goals, offer discussion.
 - **How to Answer:**
 - Clear “why this company.”
 - Stay consistent with earlier answers.
 - Compensation = researched range.
 - **Red Flags:** Negativity, perks > role, inconsistency.
 - **Examples:** Microsoft, Atlassian, Salesforce, Apple.
 - **Action:**  Prepare 3 “Why us” stories.
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9) AI Software Engineer Round

- **What:** End-to-end ML/AI engineering.
- **Format:** Coding, ML theory, design, LLM/GenAI.
- **How to Answer:**
 - Tie models to business KPIs.
 - Show experiment → measure → improve loop.
 - Discuss production pipelines + monitoring.
- **Red Flags:** Only theory or hacks, ignoring data quality.
- **Examples:** OpenAI, DeepMind, Anthropic, Microsoft Copilot, NVIDIA.
- **Action:**  Weekly ML coding + one case study (RAG, feature store).